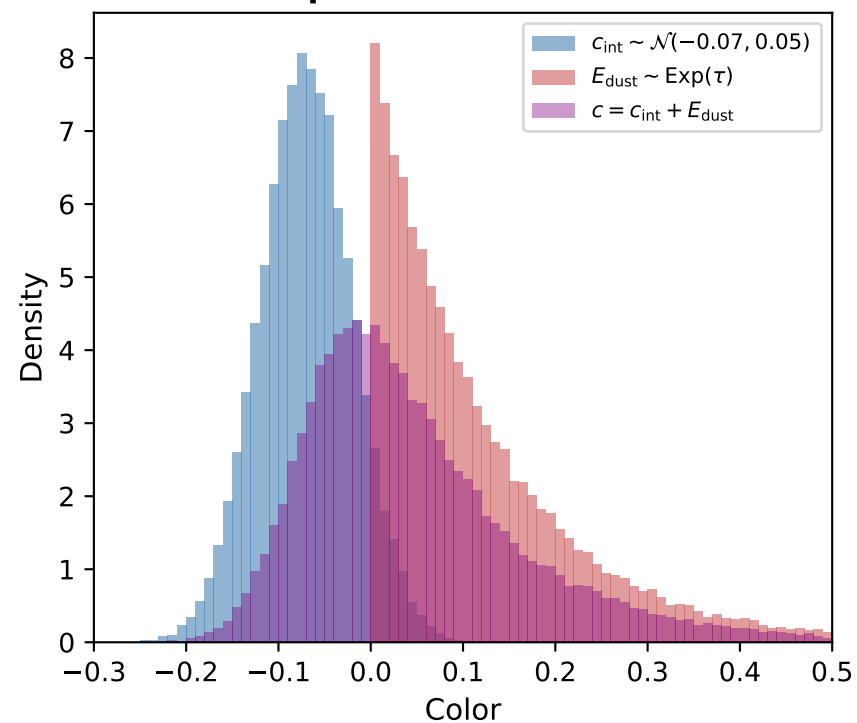
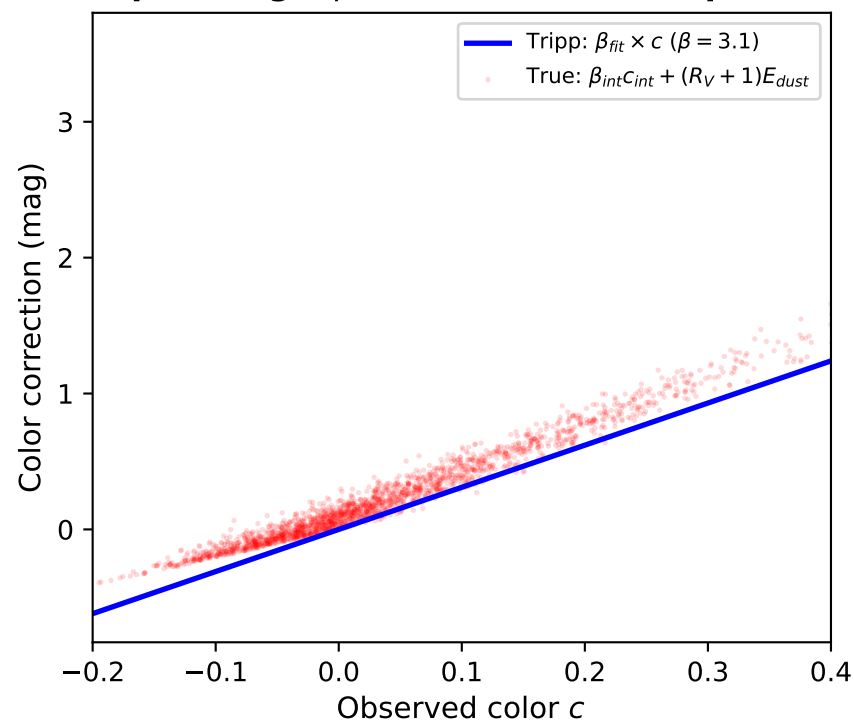


How P23 dust biases $f\sigma_8$ and how the Student- t fixes it

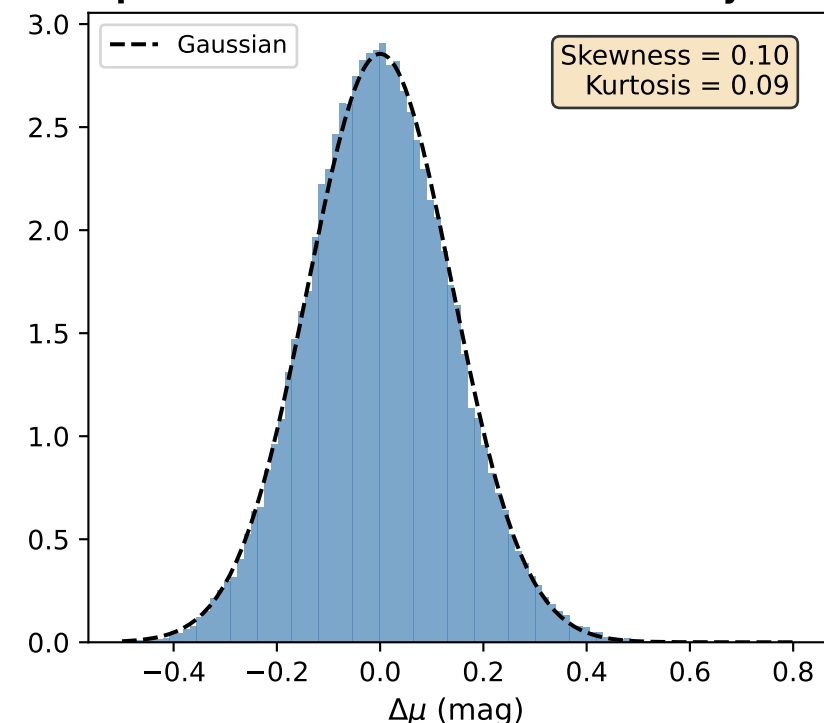
Step 1: P23 color model



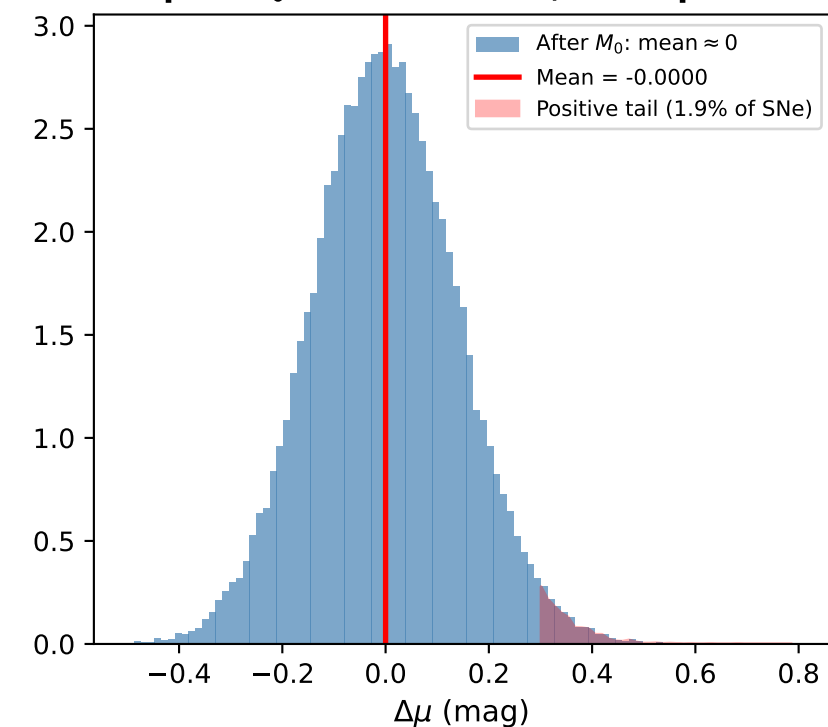
Step 2: Single β cannot fit two components



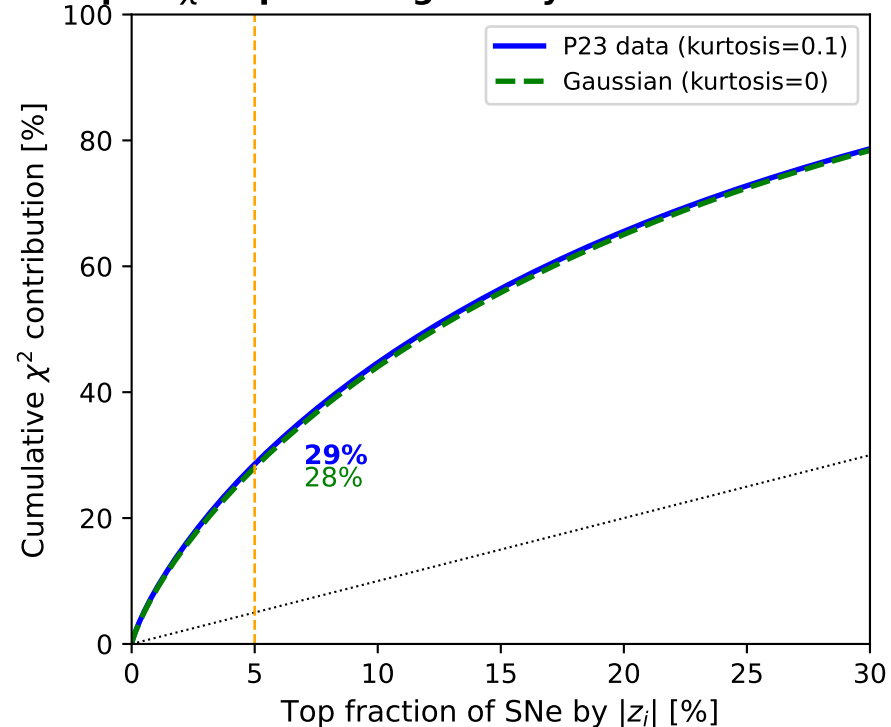
Step 3: Residuals are skewed + heavy-tailed



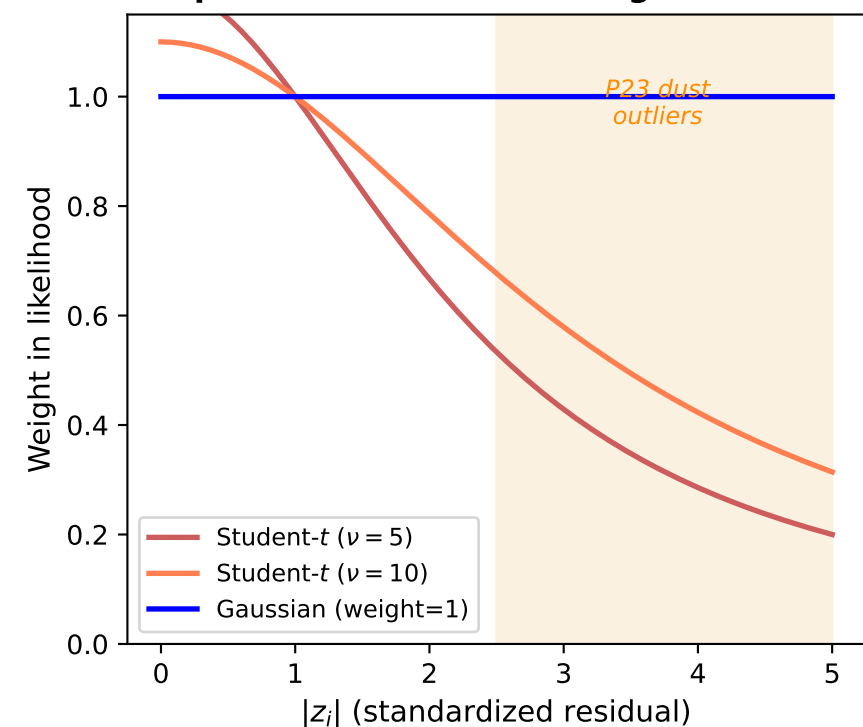
Step 4: M_0 absorbs mean, skew persists



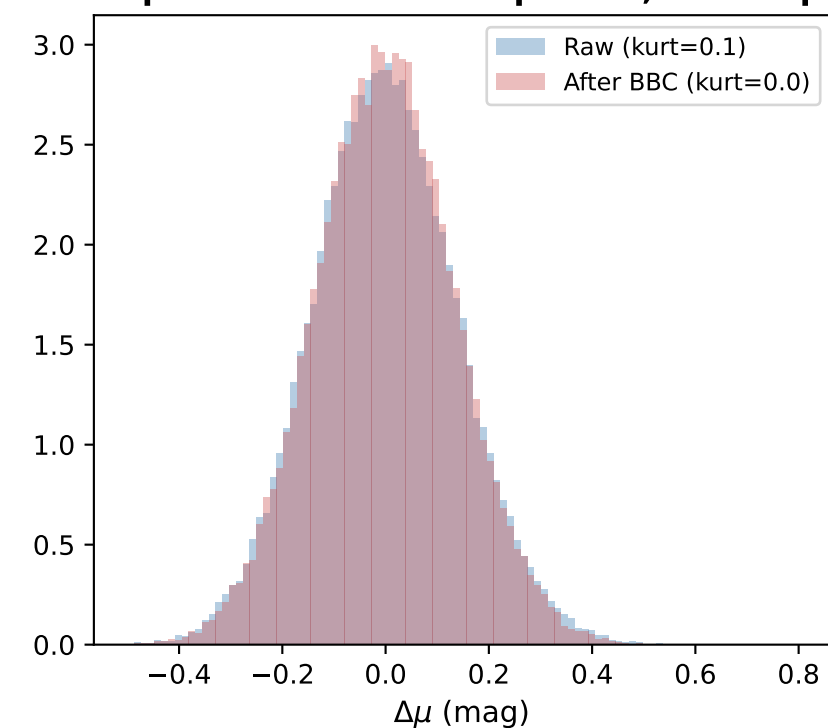
Step 5: χ^2 squares sign away $\rightarrow$ kurtosis dominates



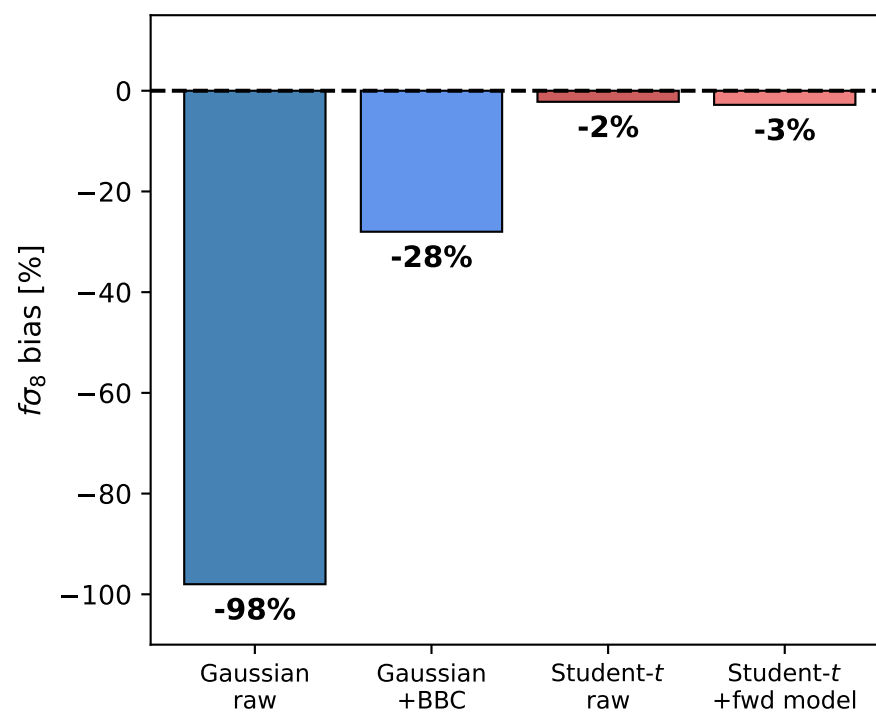
Step 6: Student- t down-weights outliers



Step 7: BBC fixes mean per bin, not shape



Result: on real SNANA P23 data



THE FULL STORY

1. P23 dust: $c = \text{Gaussian} + \text{Exponential}$
2. Single β Tripp can't fit both components
3. Residuals become skewed + heavy-tailed
4. M_0 absorbs the mean $\rightarrow$ skewness harmless
5. χ^2 squares residuals $\rightarrow$ only kurtosis matters
6. Student- t : $\log(\chi^2)$ de-leverages outliers
7. BBC fixes mean (already OK), not shape

Student- t fixes $f\sigma_8$: $-98\% \rightarrow -2\%$